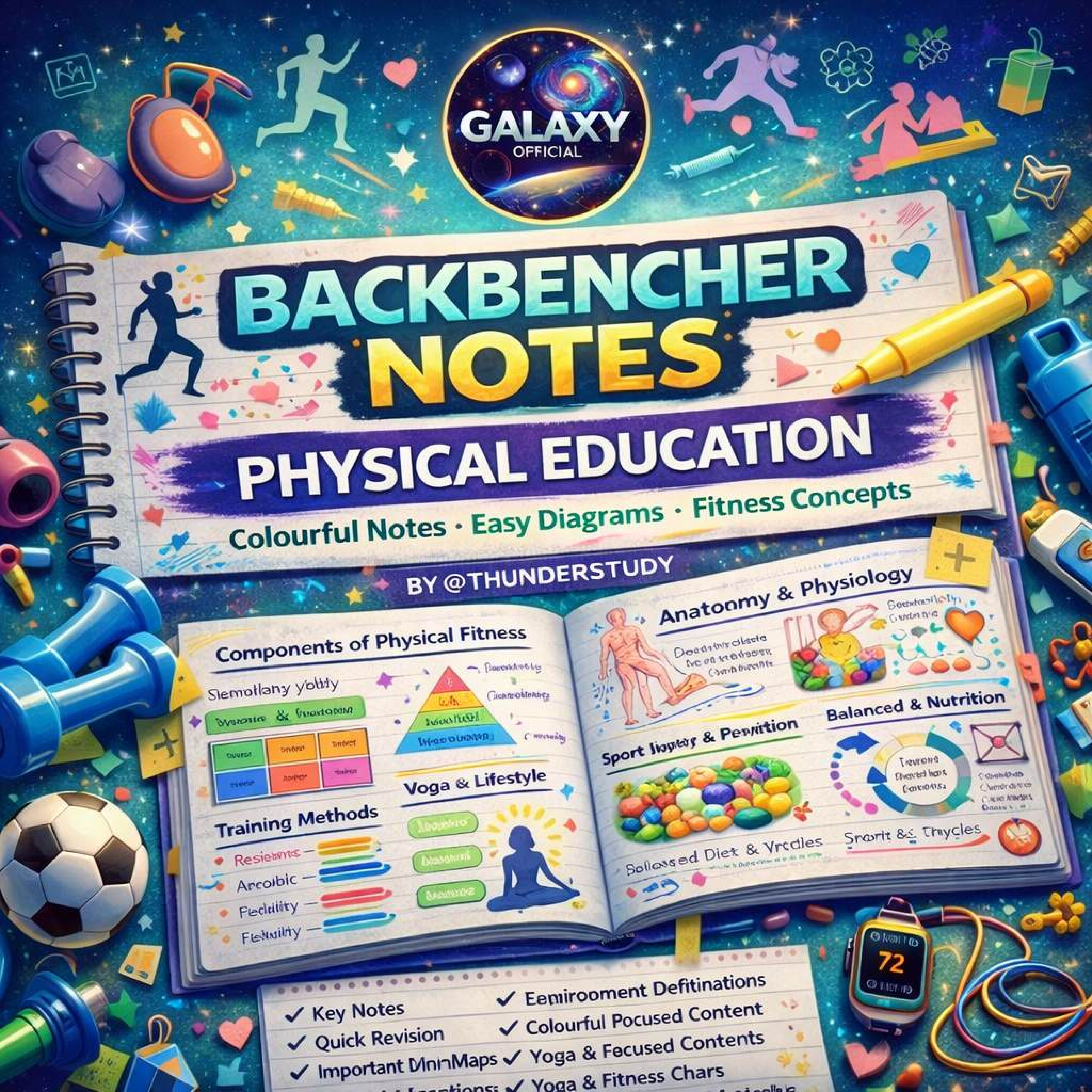

GALAXY
OFFICIAL
BACKBENCHER NOTES
PHYSICAL EDUCATION
Colourful Notes · Easy Diagrams · Fitness Concepts
BY @THUNDERSTUDY
Components of Physical Fitness
Anatomy & Physiology
Balanced & Nutrition
Yoga & Lifestyle
Training Methods
Aerobic
72
Key Notes
Quick Revision
Colourful Pocused Content
Yoga & Fecused Contents
Yoga & Fitness Chars

chapter 4

special olympics :- july 1968 in chicago
Special Olympics in indian :- 1988
Special Olympics Committee in 1988

first time :- 1968 in chicago
last time :- 2023 berlin
next time :- 2027 perth

Special Olympics

paralympic Games :- 1960 in Rome.
next time :- 2028 in los Angeles
international paralympics community was established on 22 September 1989 in bonn

Paralympics

Deaflympics :- 1924 in Paris
Deaflympic winter's game started in 1949

held once in 4 years not held twice due to World war 2
in 1966 to 1999 known as World Games for the Deaf
in 2001 this game was known as Deaflympic

Deaflympic

first time in paris in 1924 winter in seefeld in 1949
last time in caxias do sul in 2021. winter in Ankara in 2024
next time in Tokyo in 2025

World Disability Day :- 3 December

Chapter :- 6

The Milfflin Jeor Equation in 1990

senior citizen fitness test by Rikli and Jones :- 2001

Johnson - Metheny Test Battery :- 1932

Metheny Revised version of Johnson's educability test :- 1938

Harris And Benedict Revised the equation to calculate BMR :- 1984

Chapter :- 10

Periodisation of training cycle was originated in USSR in 1960s

All Formula

METHOD OF CALCULATING TEAMS IN EACH HALF

$$\text{UPPER HALF} = \frac{\text{TOTAL NUMBER OF TEAMS} + 1}{2}$$

$$\text{Also} = \frac{N + 1}{2}$$

$$\text{LOWER HALF} = \frac{\text{TOTAL NUMBER OF TEAMS} - 1}{2}$$

$$\text{Also} = \frac{N - 1}{2}$$

FORMULA TO CALCULATE NUMBERS OF BYES

TOTAL NUMBER OF TEAMS - HIGHEST POWER. = Byes

HIGHEST POWER
2
4
8
16
32
64
128

$$\text{NUMBER OF BYES IN THE UPPER HALF} = \frac{NB - 1}{2}$$

$$\text{NUMBER OF BYES IN THE LOWER HALF} = \frac{NB + 1}{2}$$

DISTRIBUTION OF BYES

1 BYES :- LOWER HALF LAST
2 BYES :- UPPER HALF FIRST
3 BYES :- LOWER HALF FIRST
4 BYES :- UPPER HALF BLAST
5 BYES :-LOWER HALF SECOND LAST

League Tournament Formulas

N = Number for Team

- Total Matches (Single League) = N (N-1) / 2
  2
- Total Matches (Double League) = N (N-1)
- Rounds (Even Teams) = N-1
- Rounds (Odd Teams) = N

FORMULA OF CALCULATING WINNER IN LEAGUE TOURNAMENT

$$\text{BRITISH METHOD} = \frac{\text{total points obtained}}{\text{total possible points}} \times 100$$

total points obtained = number of matches win + number of matches draw

WINNER 2 POINTS
LOOSER 0 POINTS
DRAW 1 POINT

$$\text{AMERICAN METHOD} = \frac{\text{matches won}}{\text{matches played}} \times 100$$

BODY MASS INDEX (BMI) = $\left(\dfrac{\text{BODY WEIGHT}}{\text{HEIGHT X HEIGHT}}\right)$

BASAL METABOLIC RATE (BMR) =

MEN = 88.362 + (13.379 + WEIGHT IN KG) + (4.739 X HEIGHT IN CM) - (5.677 X AGE IN YEAR)

WOMEN = 447.593 + (9.247 + WEIGHT IN KG) + (3.098 X HEIGHT IN CM) - (4.330 X AGE IN YEAR)

THE MILFFLIN JEOR EQUATION =

MEN = (9.99 + WEIGHT IN KG) + (6.25 X HEIGHT IN CM) - (4.92 X AGE IN YEAR) + 5

WOMEN = (9.99 + WEIGHT IN KG) + (6.25 X HEIGHT IN CM) - (5 X AGE IN YEAR) - 161

IMP THINGS

MENARCHE = FIRST PERIOD START BETWEEN 8 TO 16 YEAR OLD GIRLS
LAST FOR 2 TO 7 DAYS
NORMAL CYCLE IS OF 21 TO 35 DAYS

WHO OVERWEIGHT AND OBESITY BY BMI
UNDERWEIGHT = <18.5
NORMAL WEIGHT = 18.5 - 24.9
OVERWEIGHT = 25 - 29.9

OBESITY Class I :- 30 - 34.9
OBESITY Class II. :- 35 - 39.9
OBESITY Class III :- > 40

DIVISIONING PROCESS FOR SPECIAL OLYMPICS
DIVIDE BY MALE AND FEMALE

DIVIDE BY AGE
INDIVIDUAL
8-11 YEAR
12-15 YEAR
16-21 YEAR
22-29 YEAR
30 AND ABOVE

TEAM SPORTS
15 AND UNDER
16 -21 YEAR
22 AND ABOVE

CHAPTER 1

A. Functions of Sports Event Management

Sports event management ensures the smooth execution of tournaments at school, national, and international levels. It involves planning, organizing, staffing, directing, and controlling sporting events.

1.1 Steps to Organizing a Sports Event

A sports event requires multiple stages of preparation:

1. Formation of Organizing Committees – Assigning roles to ensure smooth coordination.
2. Establishing Objectives & Theme – Setting clear goals and defining event focus.
3. Selecting Date & Venue – Choosing an appropriate time and location.
4. Budgeting & Sponsorship – Allocating funds and securing financial support.
5. Marketing & Invitations – Promoting the event and inviting teams & officials.
6. Logistics Arrangements – Ensuring transportation, accommodation, medical assistance, and refreshments.
7. Finalizing Officials & Teams – Selecting referees, umpires, and participants.

1.2 Five Key Functions of Management

Function	Description
Planning	Setting objectives and strategies for the event.
Organizing	Arranging resources, staff, and infrastructure.
Staffing	Assigning key personnel for different roles.
Directing	Guiding and supervising staff and volunteers.
Controlling	Evaluating the event's success and making improvements.

1.3 Objectives of Planning

1. Ensures coordination and reduces errors.
2. Helps in budget control and resource allocation.
3. Promotes creativity and innovation.
4. Provides clear direction to achieve event goals.

Committee	Responsibilities
Technical Committee	Selects and manages referees, umpires, and timekeepers.
Transport Committee	Arranges travel for teams and officials.
Reception & Lodging Committee	Welcomes guests and arranges accommodations.
Medical & First Aid Committee	Provides healthcare and emergency services.
Ground & Equipment Committee	Prepares venues and arranges sports equipment.
Protest Committee	Resolves disputes and appeals.
Marketing & Publicity Committee	Handles media coverage, sponsorships, and promotions.
Finance Committee	Manages budgeting and financial transactions.

Concept	Explanation
Bye	A team advances to the next round without playing (used to balance fixtures).
Seeding	Strong teams are placed in such a way that they do not face each other early.

3.1 Knockout Tournament

- Definition: Teams play elimination matches losers exit the competition.
- Advantages:
  1. Saves time, money, and effort.
  2. Increases competitiveness.
- Disadvantages:
  1. Fewer opportunities for weaker teams.
  2. A strong team may get eliminated early due to a single loss.

3.2 League Tournament (Round Robin)

- Definition : Each team competes against every other team at least once.
- Types : a. Single League – Every team plays once against each other.
  b. Double League – Every team plays twice against each other.

Methods for League Fixtures

1. Cyclic Method – Teams rotate positions in a cycle.
2. Staircase Method – Fixtures are arranged in a ladder format.
3. Tabular Method – Matches are structured in a table format.

Intramural (Within School)

- Definition: Competitions held within a school or institution.
- Objectives:
  a. Encourages mass participation.
  b. Develops team spirit & leadership.
  c. Provides recreation & stress relief.

Extramural (Outside School)

- Definition: Competitions between different schools or institutions.
- Objectives:
  a. Enhances sports skills & experience.
  b. Improves sportsmanship and social interaction.
  c. Prepares athletes for higher-level competitions.

E. Community Sports Programs

Program	Purpose
Sports Day	Celebrates athletic achievements within schools or communities.
Health Run	Raises awareness about health and fitness.
Run for Fun	Encourages mass participation in fitness activities.
Run for Unity	Symbolizes national integration and peace.
Run for a Specific Cause	Raises awareness about social issues like cancer, education, and environment.

Chapter 2 WOMEN AND CHILD DEVELOPMENT

A. WHO Exercise Guidelines for Different Age Groups

1.1 Infancy (0-2 years)

a. Activities for head control, sitting, crawling.
b. Encouraging reaching, grasping, throwing, catching.

1.2 Early Childhood (3-6 years)

a. Focus on movement skills, coordination, and basic exercises.
b. Non-competitive and fun-based activities.

1.3 Middle Childhood (7-10 years)

a. Development of fine & gross motor skills.
b. Basic sports training and team games introduced.

1.4 Later Childhood (11-12 years)

a. Enhancing body control, coordination, and strength.
b. Team sports & social awareness through games.

1.5 Adolescence (13-18 years)

a. 60 minutes+ daily of moderate-to-vigorous physical activity.
b. Muscle & bone strengthening exercises at least 3 times a week.
c. Resistance training and endurance sports (running, swimming).

1.6 Adulthood (19-60 years)

a. Daily moderate-intensity exercise (walking, cycling).
b. Strength & resistance training at least 2 times a week.

1.7 Old Age (60+ years)

a. Light activities (walking, stretching) for 45-60 minutes, 5 days/week.
b. Climbing stairs and resistance exercises for strong bones & muscles.

B. Common Postural Deformities & Their Corrective Measures

2.1 Knock Knee (Genu Valgum)

a. Knees touch, but ankles remain apart.
b. Causes : Weak muscles, obesity, vitamin D deficiency.
c. Correction Exercises : Walking on an outward-inclined surface, horse riding.
d. Yoga : Padmasana, Gomukhasana.

2.2 Flat Foot (Pes Planus)

a. Absence of foot arch, entire sole touches the ground.
b. Causes : Poor posture, prolonged standing, weak muscles.
c. Correction Exercises : Walking on toes, skipping, picking objects with toes.
d. Yoga : Adho Mukhasana, Vajrasana.

2.3 Bow Legs (Genu Varum)

a. Legs curve outward at the knees, ankles touch.
b. Causes : Lack of calcium, vitamin D deficiency.
c. Correction : Braces, calcium-rich diet, walking on inner edges of feet.

2.4 Round Shoulders

a. Shoulders droop forward, head tilts downward.
b. Causes : Poor posture, carrying heavy loads, weak back muscles.
c. Correction Exercises : Chest expansion, reverse shoulder stretch, planks.
d. Yoga : Chakrasana, Dhanurasana.

2.5 Kyphosis (Hunchback)

a. Excessive forward curvature of the upper spine.
b. Causes : Poor posture, weak back muscles, osteoporosis.
c. Correction Exercises : Strengthening exercises, swimming.
d. Yoga : Bhujangasana, Chakrasana.

2.6 Lordosis (Swayback)

a. Excessive inward curve in the lower spine.
b. Causes: Weak abdominal muscles, obesity, poor posture.
c. Correction Exercises : Sit-ups, back-strengthening movements.
d. Yoga : Halasana, Dhanurasana.

2.7 Scoliosis

a. Sideways curvature of the spine (C or S-shaped).
b. Causes: Genetic factors, poor sitting posture, muscle imbalance.
c. Correction Exercises : Hanging from a bar, swimming (breaststroke).
d. Yoga : Trikonasana, Adho Mukhasana.

C. Women's Participation in Sports

3.1 Benefits of Sports for Women

1. Physical Benefits:

a. Improves fitness, flexibility, and bone health.
b. Reduces risk of osteoporosis and obesity.

2. Psychological Benefits:

a. Boosts self-esteem and mental resilience.
b. Reduces stress, anxiety, and depression.

3. Social Benefits:

a. Encourages teamwork, leadership, and equality.

3.2 Milestones in Indian Women's Sports

a. 1952 Olympics : First Indian woman participated.
b. 2000 Olympics : Karnam Malleswari (weightlifting) won India's first female Olympic medal (Bronze).
c. Recent Achievements : PV Sindhu (Badminton), Mary Kom (Boxing), Mirabai Chanu (Weightlifting), Lovlina Borgohain (Boxing).

D. Special Considerations for Women in Sports

4.1 Menarche & Menstrual Dysfunction

• **Menarche:** The first menstrual cycle in adolescent girls.

• **Menstrual Dysfunction:** Irregular or painful periods affecting performance.

• Causes: Stress, poor diet, excessive training, hormonal imbalance.

• Common Disorders:

a. Primary Amenorrhea: Absence of menstruation at puberty.
b. Secondary Amenorrhea: Missing periods for 3+ months.

4.2 Effects of Menstrual Health on Sports Performance

a. Low energy, weakness, mood swings, dehydration.
b. Proper diet & hydration help manage menstrual health.

E. Female Athlete Triad

A syndrome affecting women athletes due to extreme training and poor nutrition.

5.1 Components of Female Athlete Triad

1. Disordered Eating:

- Extreme dieting, calorie restriction, or eating disorders.

2. Amenorrhea:

- Irregular or absent menstrual cycles due to low energy intake.

3. Osteoporosis:

- Weak bones due to calcium loss, leading to fractures.

5.2 Causes of Female Athlete Triad

a. Pressure to maintain low body weight.
b. Excessive exercise with poor nutrition.
c. Sports like gymnastics, distance running, wrestling, and figure skating increase risk.

5.3 Prevention & Treatment

a. Balanced diet with calcium, protein, and vitamins.
b. Adequate rest & recovery periods.
c. Medical checkups & counseling.

Chapter 3 : Yoga.

A. Introduction to Yoga

a. Meaning : The word Yoga is derived from the Sanskrit word Yuj, meaning union (of body, mind, and soul).
b. Purpose : Enhances physical, mental, and spiritual well-being.
c. International Yoga Day : Celebrated on June 21 (recognized by the UN in 2015).

Elements of Yoga (Ashtanga Yoga by Patanjali)

1. Yama (Self-restraint)
2. Niyama (Self-discipline)
3. Asana (Postures)
4. Pranayama (Breathing control)
5. Pratyahara (Withdrawal of senses)
6. Dharana (Concentration)
7. Dhyana (Meditation)
8. Samadhi (Ultimate awareness)

B. Yoga for Obesity

What is Obesity?

a. Excess fat accumulation with BMI > 30.
b. Causes : Overeating, lack of exercise, genetics, high-carb diet, medications, stress.
c. Health Risks : Diabetes, hypertension, heart disease, arthritis.

Asanas for Obesity -

1. Tadasana (Mountain Pose) – Improves posture, increases height.
2. Katichakrasana (Standing Twist Pose) – Aids digestion, reduces belly fat.
3. Pavanmuktasana (Wind-Relieving Pose) – Helps relieve bloating and gas.
4. Matsyasana (Fish Pose) – Enhances metabolism, stretches abdomen.
5. Halasana (Plow Pose) – Strengthens the back and reduces belly fat.
6. Paschimottanasana (Seated Forward Bend) – Enhances digestion, tones abdomen.
7. Ardha Matsyendrasana (Half Spinal Twist) – Stimulates metabolism.
8. Dhanurasana (Bow Pose) – Stretches and tones the abdominal area.
9. Ushtrasana (Camel Pose) – Reduces fat, improves digestion.
10. Suryabhedan Pranayama – Increases energy, burns fat.

C. Yoga for Diabetes

What is Diabetes?

i. A metabolic disorder causing high blood sugar levels due to:
   a. Type 1 Diabetes: Body does not produce insulin.
   b. Type 2 Diabetes: Body produces insulin but does not use it efficiently.
ii. Causes: Sedentary lifestyle, obesity, stress, poor diet.
iii. Symptoms: Fatigue, frequent urination, excessive thirst, weight loss, blurred vision.

Asanas for Diabetes -

1. Katichakrasana – Regulates insulin.
2. Pavanmuktasana – Stimulates pancreas.
3. Bhujangasana (Cobra Pose) – Activates pancreas.
4. Shalabhasana (Locust Pose) – Improves metabolism.
5. Dhanurasana (Bow Pose) – Strengthens the pancreas.
6. Supta Vajrasana – Enhances digestion.
7. Paschimottanasana – Stimulates internal organs.
8. Ardha Matsyendrasana – Helps control blood sugar levels.
9. Mandukasana (Frog Pose) – Increases insulin secretion.
10. Gomukhasana (Cow Face Pose) – Balances sugar levels.
11. Yogmudra – Improves digestion and blood circulation.
12. Ushtrasana (Camel Pose) – Reduces stress and improves digestion.
13. Kapalabhati (Breathing Exercise) – Enhances oxygen supply and regulates sugar levels.

D. Yoga for Asthma

What is Asthma?

a. Chronic lung disease causing difficulty in breathing due to inflamed airways.
b. Causes: Dust, smoke, air pollution, allergens.
c. Symptoms: Shortness of breath, wheezing, chest tightness, coughing.

Asanas for Asthma

1. Tadasana (Mountain Pose) – Expands lungs.
2. Urdhwa Hastottanasana – Strengthens respiratory muscles.
3. Uttan Mandukasana – Opens the chest.
4. Bhujangasana (Cobra Pose) – Improves lung function.
5. Dhanurasana (Bow Pose) – Expands chest cavity.
6. Ushtrasana (Camel Pose) – Enhances lung capacity.
7. Vakrasana (Twisting Pose) – Increases oxygen intake.
8. Kapalabhati (Breathing Exercise) – Cleanses lungs.
9. Gomukhasana (Cow Face Pose) – Strengthens lungs.
10. Matsyasana (Fish Pose) – Improves airflow in lungs.
11. Anulom-Vilom (Alternate Nostril Breathing) – Improves breathing efficiency.

E. Yoga for Hypertension (High Blood Pressure)

What is Hypertension?

a. A condition where blood pressure is persistently high (>140/90 mmHg).
b. Causes: Stress, obesity, smoking, alcohol, lack of exercise.
c. Symptoms: Headache, dizziness, fatigue, vision problems.

Asanas for Hypertension

1. Tadasana – Reduces stress.
2. Katichakrasana – Lowers blood pressure.
3. Uttanpadasana – Improves circulation.
4. Ardha Halasana – Relaxes body.
5. Saral Matsyasana – Enhances oxygen supply.
6. Gomukhasana – Relaxes nervous system.
7. Uttan Mandukasana – Reduces stress.
8. Vakrasana – Improves flexibility.
9. Bhujangasana (Cobra Pose) – Reduces anxiety.
10. Makarasana (Crocodile Pose) – Relieves stress.
11. Shavasana (Corpse Pose) – Deep relaxation.
12. Nadi Shodhana Pranayama – Calms the mind.
13. Sitli Pranayama – Lowers body temperature.

F. Yoga for Back Pain & Arthritis

6.1 What is Back Pain?

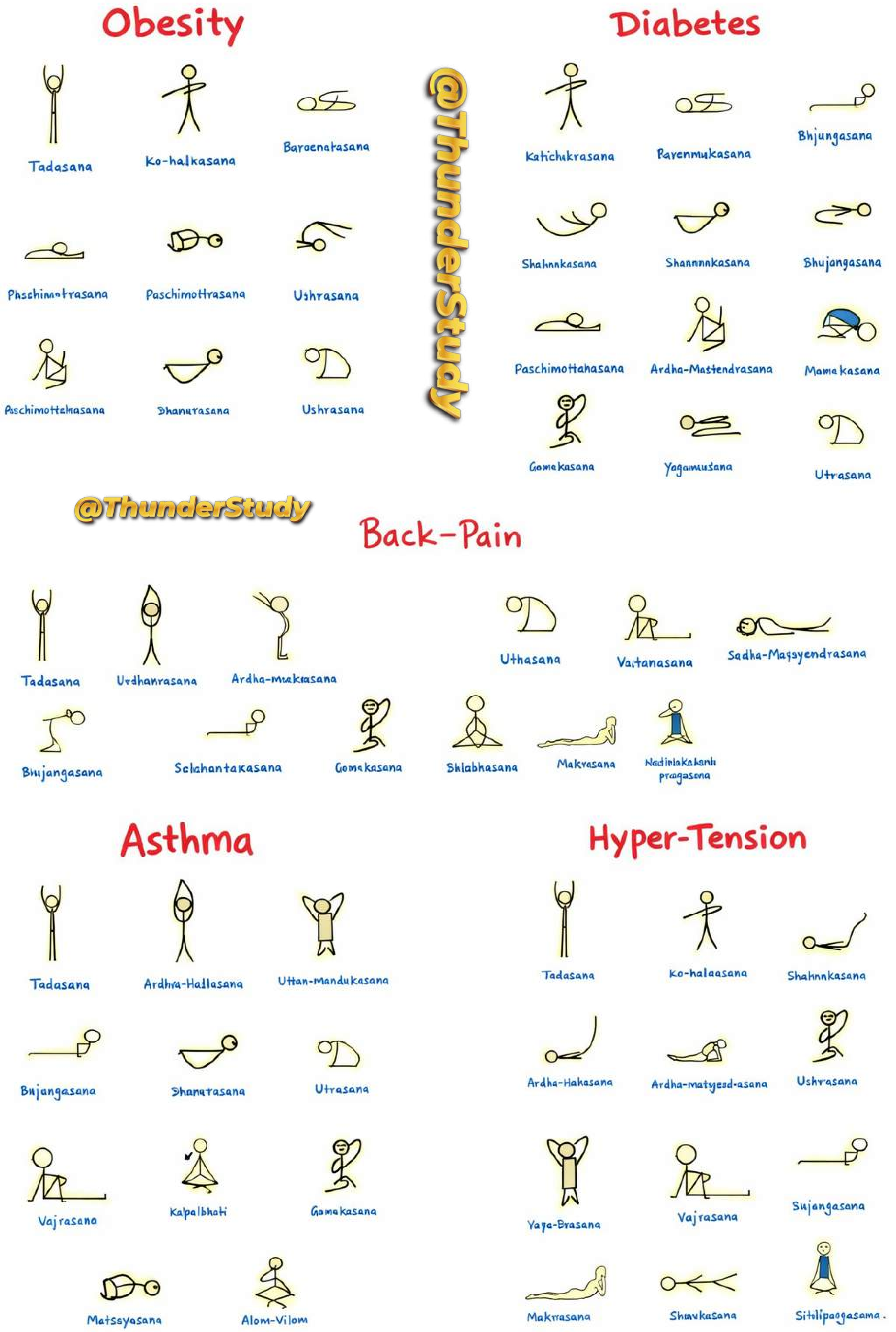
a. Pain in the spine or lower back due to muscle strain, injury, or poor posture.
b. Causes: Overweight, lack of exercise, poor posture, stress.

6.2 What is Arthritis?

a. Inflammation of the joints causing pain, stiffness, and swelling.
b. Causes: Aging, genetic factors, obesity.

Asanas for Back Pain & Arthritis

1. Tadasana – Improves posture.
2. Urdhawa Hastottanasana – Stretches back muscles.
3. Ardha Chakrasana (Half Wheel Pose) – Reduces stiffness.
4. Ushtrasana – Strengthens spine.
5. Vakrasana – Improves flexibility.
6. Saral Matsyendrasana – Eases back tension.
7. Bhujangasana (Cobra Pose) – Strengthens back muscles.
8. Gomukhasana – Reduces joint stiffness.
9. Bhadrasana (Butterfly Pose) – Improves hip flexibility.
10. Makarasana (Crocodile Pose) – Relieves back pain.
11. Nadi Shodhana Pranayama – Reduces inflammation.


Obesity
Tadasana
Ko-halkasana
Baroenatasana
Phschimotrasana
Paschimottrasana
Ushrasana
Paschimottahasana
Dhanurasana
Ushrasana
Diabetes
Katichakrasana
Pavenmukasana
Bhjungasana
Shahnnkasana
Shannnnkasana
Bhujangasana
Paschimottahasana
Ardha-Mastendrasana
Mamakasana
Gomakasana
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Back-Pain
Tadasana
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Ardha-mrakrasana
Uthasana
Vaitanasana
Sadha-Matsyendrasana
Bhujangasana
Sclahantakasana
Gomakasana
Shlabhasana
Makrasana
Nadinlakahanh pragasona
Asthma
Tadasana
Ardhva-Hallasana
Uttan-Mandukasana
Bujangasana
Dhanurasana
Utrasana
Vajrasano
Kapalbhati
Gomakasana
Matssyasana
Alom-Vilom
Hyper-Tension
Tadasana
Ko-halaasana
Shahnnkasana
Ardha-Hakasana
Ardha-matsyend-asana
Ushrasana
Yaga-Brasana
Vajrasana
Sujangasana
Makrrasana
Shavukasana
Sitilipaagasama.

Chapter 4 :- PHYSICAL EDUCATION AND SPORTS FOR CWSN

A. Concept of Disability & Disorder

1.1 What is Disability?

a. A disability is a long-term impairment that affects a person's ability to perform daily activities such as walking, eating, or communicating.
b. It can be physical, intellectual, sensory, or mental in nature.
c. The Rights of Persons with Disabilities (RPWD) Act, 2016 recognizes and protects the rights of disabled individuals in India.

Types of Disabilities (As per RPWD Act, 2016)

1. Physical Disability – Loss of body function or structure (e.g., paralysis, amputations).
2. Intellectual Disability – Cognitive impairments (e.g., Down syndrome, autism).
3. Sensory Disability – Affects senses like vision or hearing (e.g., blindness, deafness).
4. Mental Disability – Affects psychological well-being (e.g., depression).

1.2 What is a Disorder?

a. A disorder is a medical condition that affects a person's normal functioning, often linked to mental health, behavior, or body function.
b. Example: ADHD, dyslexia, speech disorders.

Difference Between Disability & Disorder

Disability	Disorder
Long-term impairment	Temporary or chronic medical condition
Can be physical, intellectual, or sensory	Often mental or behavioral
Recognized under RPWD Act	More common in medical diagnosis

F. Benefits of Physical Activities for CWSN

6.1 Physical Benefits

1. Improves muscle strength and coordination.
2. Reduces obesity, high blood pressure, and diabetes risk.
3. Enhances balance and flexibility.

6.2 Psychological Benefits

4. Boosts self-confidence and self-esteem.
5. Reduces anxiety, depression, and stress.
6. Improves cognitive skills like reasoning and memory.

6.3 Social Benefits

7. Encourages teamwork and cooperation.
8. Enhances communication and friendship-building.

G. Strategies to Make Physical Activities Accessible for CWSN

1. Medical Check-Up : Identify specific disabilities and design appropriate programs.

2. Interest-Based Activities : Select activities based on a child's preferences, abilities, and past experience.

3. Use of Different Teaching Strategies : Incorporate visual, verbal, and peer-assisted learning to accommodate diverse needs.

4. Modification of Rules : Adjust rules (e.g., extra time, fewer opponents) to allow fair participation.

5. Creating a Suitable Environment : Provide quiet zones for children with autism and use wheelchair-accessible sports areas.

B. Organizations Promoting Disability Sports

2.1 Special Olympics

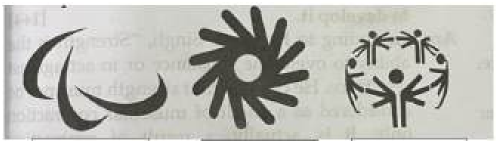
a. Global sports organization for individuals with intellectual disabilities.
b. Founded in 1968 in Chicago, USA.
c. Motto: "Let me win. But if I cannot win, let me be brave in the attempt."
d. Events: Held every four years, alternating between Summer and Winter Games.

2.2 Paralympics

a. International multi-sport event for athletes with physical disabilities.
b. First held in 1960 in Rome, Italy.
c. Governed by: International Paralympic Committee (IPC).
d. Motto: "Spirit in Motion."
e. India's Performance:
f. First participated in 1968.
g. Best finish in 2020 (19 medals: 5 Gold, 8 Silver, 6 Bronze).
h. 2024 Paralympics to be held in Paris, France.

2.3 Deaflympics

a. Sports event for hearing-impaired athletes.
b. Started in 1924 in Paris, France.
c. Governed by: International Committee of Sports for the Deaf (ICSD).
d. Motto: "Equality through sports."
e. Special Features:
f. No whistles, starting guns, or verbal commands; uses lights, flags, and hand signals.
g. Hearing aids and cochlear implants are not allowed during competitions.



Paralympics | Deaflympics | Special Olympics

C. Classification & Divisioning in Sports

3.1 What is Classification?

a. Classification ensures fair competition by grouping athletes based on functional ability or impairment, rather than skill level.
b. Used in: Paralympics, Deaflympics, and Special Olympics.
c. Example: In Paralympics, wheelchair racers compete against others with similar disabilities.

3.2 What is Divisioning?

a. Divisioning further categorizes athletes based on their skill level within their classification.
b. Example: In Special Olympics, athletes with intellectual disabilities are grouped into divisions based on performance ability rather than disability type.

How Classification & Divisioning Work

1. Medical Examination – Evaluates disability type.
2. Functional Testing – Assesses movement, coordination, and skill level.
3. Assignment to Class or Division – Ensures fair competition.

D. Inclusion in Sports: Concept, Need & Implementation

4.1 What is Inclusion?

a. Inclusion means providing equal opportunities for CWSN (Children with Special Needs) to participate in sports alongside able-bodied individuals.
b. Ensures accessibility, social integration, and personal development.

4.2 Need for Inclusion in Sports

1. Encourages social skills and teamwork.
2. Develops a sense of belonging and equality.
3. Improves motor skills and coordination.
4. Boosts self-confidence and mental well-being.

4.3 How to Implement Inclusion in Schools

1. Collaboration between teachers, parents, and therapists.
2. Trained staff for CWSN.
3. Modified equipment and rules to accommodate disabilities.
4. Use of adaptive technology (e.g., braille scorecards, prosthetics).

E. Key Professionals Supporting Inclusive Physical Education

Professional	Role
School Counsellor	Helps CWSN in academics, social, and emotional development.
Physical Education Teacher	Designs adaptive sports activities.
Speech Therapist	Assists children with speech and communication difficulties.
Special Education Teacher	Supports CWSN in classroom learning.

Chapter 5 :- FOOD AND NUTRITION

A. Balanced Diet and Nutrition

1.1 What is Balanced Diet?

A balanced diet contains the right amount of essential nutrients (carbohydrates, proteins, fats, vitamins, minerals, and water) to support body functions.

It helps in:

- Growth and development
- Energy production
- Tissue repair
- Immune system support

1.2 What is Nutrition?

Nutrition is the process of obtaining and consuming food for energy and body maintenance. Proper nutrition helps athletes stay hydrated, fuel their bodies, and improve performance.

Goals of Nutrition in Sports

a. Hydration – Prevents dehydration and maintains energy levels.
b. Fuelling the body – Provides energy for training and competitions.
c. Performance enhancement – Boosts strength, endurance, and recovery.
d. Muscle preservation – Prevents muscle breakdown.
e. Recovery improvement – Aids in post-exercise muscle repair.

D. Eating for Weight Control

Maintaining a healthy weight depends on calorie intake vs. calorie expenditure.

4.1 BMI Calculation

• Formula : $\boxed{\text{BMI} = \text{Weight (kg)} / \text{Height (m)}^2}$

a. Underweight : BMI < 18.5
b. Normal Weight : BMI 18.6-24.9
c. Overweight : BMI 25-29.9
d. Obesity : BMI ≥ 30

4.2 Pitfalls of Dieting

a. Skipping meals – Slows metabolism.
b. Extreme calorie reduction – Causes weakness.
c. Over-reliance on diet pills – Harmful side effects.
d. Low-fat diets – Can lead to vitamin deficiencies.

4.3 Food Intolerance & Myths

a. Food Intolerance: Difficulty in digesting certain foods (e.g., lactose intolerance).
b. Common Food Myths:
c. Eggs increase cholesterol – Not true; they are a good protein source.
d. Eating at night makes you fat – Weight gain depends on total calorie intake.

B. Macro and Micro Nutrients

2.1 Macronutrients (Required in large amounts)

1. Carbohydrates (55-60% of diet)

• Primary energy source for the body.

• Types:

1. Simple Carbohydrates: Quick energy (e.g., sugar, honey, fruits).
2. Complex Carbohydrates: Slow-release energy (e.g., whole grains, rice, pasta, oats)

• Stored as glycogen in muscles and liver.

2. Proteins (10-15% of diet)

a. Essential for muscle repair and growth.
b. Sources: Meat, fish, eggs, dairy, legumes, nuts.
c. Recommended Intake:
d. Normal individuals: 1g/kg body weight
e. Athletes: 1.2-2g/kg body weight

3. Fats (20-30% of diet)

• Provide long-term energy and help absorb vitamins.

• Types:

a. Saturated Fats: Found in meat, dairy, butter (should be limited).
b. Unsaturated Fats: Found in nuts, fish, olive oil (healthy fats).

• 1 gram of fat = 9 kcal of energy (highest energy source).

4. Water

a. Regulates body temperature and removes toxins.
b. Prevents dehydration during exercise.

2.2 Micronutrients (Required in small amounts)

Micronutrients are essential for immune function, bone strength, and metabolism.

1. Vitamins:

- Fat-Soluble (Stored in fat): A, D, E, K
- Water-Soluble (Not stored): B-complex, C
-

Vitamin	Function	Sources
A (Retinol)	Vision, skin health	Carrots, eggs, dairy
B1 (Thiamine)	Energy production	Whole grains, nuts
B2 (Riboflavin)	Metabolism	Milk, leafy greens
C (Ascorbic Acid)	Immunity, wound healing	Citrus fruits, tomatoes
D (Calciferol)	Bone health	Sunlight, fish, eggs
E (Tocopherol)	Antioxidant	Nuts, seeds, vegetable oil
K (Phylloquinone)	Blood clotting	Green leafy vegetables

2. Minerals:

• Required for strong bones, muscle function, and nerve transmission.

Mineral	Function	Sources
Calcium	Bone & teeth formation	Milk, cheese, spinach
Iron	Haemoglobin formation	Meat, spinach, eggs
Sodium	Nerve & muscle function	Table salt, processed food
Potassium	Maintains fluid balance	Bananas, potatoes
Magnesium	Muscle & nerve function	Nuts, beans, grains

C. Nutritive & Non-Nutritive Components of Diet

3.1 Nutritive Components

a. Carbohydrates – Provide quick energy.
b. Proteins – Help in muscle growth and repair.
c. Fats – Store energy and regulate body functions.
d. Vitamins & Minerals – Support immune function, metabolism, and bone health.

3.2 Non-Nutritive Components

a. Water – Essential for digestion, circulation, and waste removal.
b. Fiber (Roughage) – Prevents constipation and aids digestion.
c. Artificial Sweeteners & Preservatives – Found in processed foods; should be limited.

E. Importance of Diet in Sports

Proper nutrition improves athletic performance and recovery.

5.1 Pre-Competition Nutrition

Goal: Maximize energy stores.

a. Meal Plan: High carbs, moderate protein, low fat.
b. Example: Whole grain bread with peanut butter, banana, and water.

5.2 During Competition Nutrition

Goal: Maintain energy and hydration.

a. Plan: Consume sports drinks, bananas, or energy bars.
b. Example: Small sips of electrolyte drinks every 15 minutes.

5.3 Post-Competition Nutrition

Goal: Recovery and muscle repair.

a. Plan: Protein-rich foods, carbohydrates, and hydration.
b. Example: Grilled chicken with rice and vegetables, water.

Chapter 6 :- TEST AND MEASUREMENT

A. Fitness Tests – SAI Khelo India Fitness Test

This test is conducted in schools to assess the physical fitness of students. It is divided into two age groups:

1.1 For Age Group 5-8 Years (Class 1-3)

At this stage, children develop Fundamental Movement Skills (FMS) such as locomotion, body management, and coordination. The test includes:

1. BMI (Body Mass Index) : Measures body composition using weight (kg) and height (m).

• Formula : $\boxed{\text{BMI = Weight (kg) / Height (m)}^2}$

• Purpose: Higher BMI indicates higher fat levels; helps track childhood obesity.

2. Flamingo Balance Test : Measures static balance by standing on one leg for 60 seconds.

• Procedure:

a. Stand on a beam.
b. Lift one leg and balance.
c. Count the number of times the balance is lost.

• Scoring: More falls = Poor balance.

3. Plate Tapping Test : Measures speed and limb coordination.

• Procedure:

a. Two discs placed 60 cm apart on a table.
b. Participant taps each disc as fast as possible for 25 cycles (50 taps).

• Scoring: Time taken to complete the cycles.

B. Measurement of Cardio-Vascular Fitness – Harvard Step Test

• Definition: Measures cardiovascular endurance by testing heart rate recovery.

• Procedure:

a. Step up and down a 45 cm bench at 30 steps per minute for 5 minutes.
b. After stopping, measure pulse rate at 1 min, 2 min, and 3 min.
c. Apply the Fitness Index formula:

$$\boxed{\text{Fitness Index = Duration of exercise (sec) x 100 / 5.5 x Pulse count after exercise}}$$

• Interpretation:

- High score = Excellent fitness.
- Low score = Poor endurance, need for training.

C. Basal Metabolic Rate (BMR)

• Definition: The minimum calories required by the body to function at rest.

• Significance:

- Higher BMR = Faster metabolism, burns more calories.
- Lower BMR = Slower metabolism, more fat storage.

• Formula (Mifflin-St Jeor Equation):

$$\text{BMR} = [\,10 \times \text{weight (kg)}\,] + [\,6.25 \times \text{height (cm)}] - [\,5 \times \text{age}\,] + S$$

S = +5 for males, -161 for females.

• How BMR Changes:

• Teenagers → Highest BMR.

• Aging → BMR slows, requiring calorie control.

E. Johnsen – Metheny Test of Motor Educability

This test evaluates neuromuscular coordination through four movements:

1. Front Roll – Measures flexibility and balance.
2. Back Roll – Assesses motor control.
3. Jumping Half Turns – Tests body coordination.
4. Jumping Full Turns (Boys Only) – Measures agility and reaction time.

1.2 For Age Group 9-18 Years (Class 4-12)

For older students, fitness assessment includes:

1. BMI (as in the younger age group).

2. 50m Speed Test : Measures sprinting speed.

• Procedure:

a. Start from a standing position.
b. Sprint 50 meters as fast as possible.

• Scoring: Time taken (in seconds).

3. 600m Run/Walk Test : Evaluates cardiovascular endurance.

• Procedure:

1. The subject runs or walks 600m at their best pace.

• Scoring: Time taken to complete the distance.

4. Sit & Reach Flexibility Test : Measures lower back and hamstring flexibility.

• Procedure:

a. Sit with legs straight and feet against a box.
b. Reach forward with hands.

• Scoring: Distance reached (in cm or inches).

5. Strength Tests:

a. Abdominal Partial Curl-Up (measures core strength).
b. Push-Ups (Boys) & Modified Push-Ups (Girls) (measure upper body strength).

D. Rikli & Jones Senior Citizen Fitness Test

Developed in 2001, this test evaluates physical fitness in older adults.

Test Components & Purpose:

1. Chair Stand Test:

a. Measures lower body strength.
b. Procedure: Sit and stand repeatedly for 30 seconds.
c. Scoring: Count of successful repetitions.

2. Arm Curl Test:

a. Measures upper body strength.
b. Procedure: Lift 5 lb (women) or 8 lb (men) weight for 30 seconds.
c. Scoring: Number of curls performed.

3. Chair Sit & Reach Test:

a. Measures lower body flexibility.
b. Procedure: While seated, reach forward to touch toes.
c. Scoring: Distance reached (in cm/inches).

4. Back Scratch Test:

a. Measures upper body flexibility.
b. Procedure: Try to touch fingers behind the back.
c. Scoring: Distance between fingertips (+ve if they touch, -ve if they don't).

5. Eight Foot Up & Go Test:

a. Measures agility and balance.
b. Procedure: Stand up, walk 8 feet, return, and sit down.
c. Scoring: Time taken.

6. Six-Minute Walk Test:

a. Measures aerobic endurance.
b. Procedure: Walk for 6 minutes; measure distance covered.

Chapter 7 :- PHYSIOLOGY AND INJURY IN SPORTS

A. Physiological Factors Determining Physical Fitness

1. Strength

a. Muscle Size: Larger muscles generate more force. Males generally have larger muscles than females.
b. Muscle Composition: Muscles have fast-twitch fibres (for explosive movements) and slow-twitch fibres (for endurance). More fast-twitch fibres result in greater strength.
c. Body Weight: Heavier individuals tend to have more strength due to larger muscle mass.

2. Flexibility

a. Joint Structure: Ball-and-socket joints (shoulders, hips) allow greater flexibility than hinge joints (knees, elbows).
b. Muscle Strength: Stronger muscles can restrict flexibility.
c. Age & Gender: Flexibility declines with age; females are naturally more flexible than males.
d. Injuries: Injuries reduce flexibility by forming less elastic scar tissue.
e. Active Lifestyle: Regular movement maintains joint flexibility, while inactivity leads to stiffness.

3. Speed

a. Explosive Strength: Quick, powerful movements depend on muscle power (e.g., sprinting, boxing).
b. Nervous System Efficiency: Faster nerve signals improve reaction time and movement speed.
c. Muscle Elasticity: Flexible muscles allow quicker contraction and movement.
d. Energy Reserves: ATP stores in muscles fuel high-speed activities.

4. Endurance

a. Aerobic Capacity: The ability of the heart and lungs to supply oxygen to muscles during prolonged activity.
b. Oxygen Uptake: Higher oxygen intake enhances endurance.
c. Cardiac Output: The heart pumps more blood to muscles, sustaining physical activity.

B. Effects of Exercise on the Cardio-Respiratory System

1. Cardio-Vascular System

a. Increased Heart Rate: More blood is pumped to deliver oxygen to muscles.
b. Increased Stroke Volume: The heart pumps more blood per beat.
c. Higher Cardiac Output: Total blood pumped per minute increases.
d. Increased Blood Flow: Active muscles receive more oxygen-rich blood.
e. Higher Blood Pressure: Temporary rise in systolic blood pressure during exercise.

2. Respiratory System

a. Increased Oxygen Intake: Lungs expand to take in more oxygen.
b. Improved Alveolar Function: More oxygen is absorbed into the bloodstream.
c. Higher Vital Capacity: The maximum amount of air inhaled and exhaled increases.
d. Residual Air Volume Expands: More air remains in the lungs, improving breathing efficiency.

C. Effects of Exercise on the Muscular System

1. Muscle Growth (Hypertrophy): Regular exercise increases muscle fibre size.
2. Improved Reaction Time: Faster muscle contractions enhance sports performance.
3. Capillarization: More blood vessels form in muscles, improving oxygen supply.
4. Fat Reduction: Exercise burns fat and maintains muscle definition.
5. Delayed Fatigue: Increased energy reserves prevent early exhaustion.

D. Physiological Changes Due to Aging

a. Reduced Muscle Strength & Endurance: Muscle fibers shrink, making movement slower
b. Decreased Lung Function: Oxygen intake efficiency declines.
c. Increased Blood Pressure: Arteries lose flexibility, raising the risk of heart issues.
d. Lower Bone Density: Weaker bones increase the risk of fractures.
e. Higher Fat Accumulation: Metabolism slows, leading to weight gain.

E. Sports Injuries & Their Management

1. Soft Tissue Injuries

a. Abrasion: Skin damage due to friction (e.g., falling on a rough surface).
b. Contusion (Bruise): Blood vessels break under the skin due to impact.
c. Laceration: Deep, irregular wounds caused by blunt trauma.
d. Sprain: Ligament overstretching (e.g., ankle twists).
e. Strain: Torn or overstretched muscle fibers (e.g., hamstring injury).

2. Bone & Joint Injuries

2.1 Dislocation : Bone moves out of its normal joint position (e.g., shoulder dislocation).

2.2 Fractures :

a. Greenstick Fracture: Partial break in children's soft bones.
b. Transverse Fracture: A straight break across the bone.
c. Oblique Fracture: Diagonal break due to angled force.
d. Stress Fracture: Tiny cracks in bones from repeated stress (e.g., running injuries).
e. Comminuted Fracture: Bone breaks into multiple fragments.

3. Prevention of Sports Injuries

a. Warm-up & Stretching: Prepares muscles and joints for physical activity.
b. Proper Training & Conditioning: Gradually increasing exercise intensity reduces injury risk.
c. Correct Techniques & Equipment: Ensures safe movements and protects the body.
d. Adequate Rest & Recovery: Prevents overuse injuries.
e. Medical Checkups & Injury Management: Detects and treats injuries early.

Chapter 8 :- **Biomechanics and Sports**

A. Meaning of Biomechanics

i. **Biomechanics** = Bio (Living organism) + Mechanics (Study of forces on a body).
ii. It is the study of how forces affect the movement and performance of the human body in sports.

B. Newton's Laws of Motion & Their Application in Sports

1. First Law (Law of Inertia):

i. An object remains at rest or in motion unless acted upon by an external force.
ii. Example: A football stops due to friction, a basketball follows a parabolic path when shot.

2. Second Law (Law of Acceleration):

i. Force = Mass × Acceleration (F = ma).
ii. Example: A baseball hit with double force will accelerate twice as fast.

3. Third Law (Law of Action & Reaction):

i. Every action has an equal and opposite reaction.
ii. Example: A swimmer pushes water backward, and water pushes them forward.

C. Types of Levers & Their Application in Sports

• A lever consists of a rigid structure (bone), force (muscle), a fulcrum (joint), and load (weight).

• **Types of Levers:**

1. First-Class Lever: Fulcrum in the middle (e.g., lifting head to head a football).

2. Second-Class Lever: Load in the middle (e.g., standing on tiptoes for a jump).

3. Third-Class Lever: Effort in the middle (e.g., biceps curl).

D. Equilibrium & Centre of Gravity

1. Equilibrium : A state of balance where opposing forces cancel out.

a. **Dynamic Equilibrium**: Balance while in motion (e.g., cartwheel in gymnastics).
b. **Static Equilibrium:** Balance while stationary (e.g., standing still).

2. Centre of Gravity : The point where an object's weight is evenly distributed.

Lower centre of gravity = Greater stability.

Example: A sprinter's centre of gravity is forward to maximize speed.

E. Friction & Its Role in Sports

• **Friction:** The force that resists motion between two surfaces.

• **Types of Friction:**

1. Static Friction: Prevents objects from moving (e.g., a skier on snow).
2. Sliding Friction: Resistance between sliding objects (e.g., ice skating).
3. Rolling Friction: Resistance for rolling objects (e.g., bicycle wheels).
4. Fluid Friction: Resistance in liquids or gases (e.g., swimming).

F. Projectile Motion in Sports

• Projectile: Any object thrown into space affected only by gravity.

• **Factors Affecting Projectile Motion:**

1. Gravity (pulls objects down).
2. Air Resistance (affects trajectory).
3. Speed of Release (higher speed = longer range).
4. Angle of Release (maximum range at 45°).
5. Height of Release (higher release = longer flight).

• Examples: Throwing a javelin, shooting a basketball.

Chapter 9 psychology and sport

A. Personality: Definition & Types

i. Personality refers to the combination of physical, mental, emotional, and behavioural traits that define an individual.
ii. Derived from the Latin word Persona, meaning "mask."

Jung's Classification of Personality -

1. **Introvert –** Shy, reserved, prefers solitude, low self-confidence.
2. **Extrovert** – Outgoing, sociable, confident, enjoys social interaction.
3. **Ambivert** – A mix of both introvert and extrovert traits.

Big Five Personality Traits

1. Openness – Creativity, curiosity, adaptability.
2. Conscientiousness – Discipline, organization, hard work.
3. Extraversion – Social, energetic, talkative.
4. Agreeableness – Kindness, cooperation, empathy.
5. Neuroticism – Emotional stability, ability to handle stress.

B. Motivation: Types & Techniques

• Definition: The internal or external force that drives an individual to achieve a goal.

• **Types of Motivation:**

1. **Intrinsic Motivation** – Comes from within (e.g., personal satisfaction, enjoyment).
2. **Extrinsic Motivation** – Driven by external rewards (e.g., money, medals, praise).

Techniques to Enhance Motivation -

a. Setting realistic goals
b. Providing rewards and recognition
c. Encouraging self-motivation and discipline
d. Offering constructive feedback

C. Exercise Adherence: Reasons, Benefits & Strategies

• Exercise adherence refers to the consistency in following an exercise routine.

• **Reasons for Exercise Adherence:**

a. Overcoming social physique anxiety
b. Reducing disease risk
c. Mental relaxation
d. Recreation and socialization

Benefits of Exercise

a. Improves overall health
b. Reduces stress and enhances happiness
c. Boosts self-efficacy and confidence
d. Encourages social interaction

Strategies to Enhance Exercise Adherence

1. Goal Setting – Setting clear, achievable fitness goals.
2. Variety in Exercise – Reducing boredom with different activities.
3. Social Support – Exercising with friends or in groups.
4. Incentives & Rewards – Using motivational reinforcements.
5. Health Education – Raising awareness about fitness benefits.

D. Aggression in Sports

• Definition: Behaviour aimed at harming others physically or psychologically.

• **Types of Aggression:**

1. **Hostile Aggression** – Intentionally causing harm out of anger.
2. **Instrumental Aggression** – Aggression used to achieve a goal (e.g., tackling in football).
3. **Assertive Behaviour** – Aggressive play within the rules of the game.

E. Psychological Attributes in Sports

1. Self-Esteem

i. An athlete's confidence in their abilities and self-worth.
ii. Higher self-esteem leads to better performance.

2. Mental Imagery

i. Visualization technique where athletes imagine themselves performing successfully.
ii. Helps improve focus and reduce anxiety before competitions.

3. Self-Talk

i. Internal dialogue that influences motivation and confidence.
ii. Positive self-talk enhances performance, while negative self-talk can reduce it.

4. Goal Setting

i. Developing a step-by-step plan to achieve desired outcomes in sports.
ii. Helps athletes stay focused and motivated.

Chapter 10 :- TRAINING IN SPORTS

A. Talent Identification and Development in Sports

Talent Identification :

Talent identification is the process of selecting athletes who have the potential to excel in sports. It involves predicting an individual's future performance based on various attributes:

1. Physical attributes – Strength, speed, endurance, flexibility.
2. Technical skills – Sport-specific movements and techniques.
3. Psychological skills – Motivation, confidence, resilience.
4. Cognitive skills – Decision-making, game sense, strategic thinking.
5. Social skills – Teamwork, leadership, and adaptability.

Talent Development :

Once athletes are identified, they must be provided with proper training and support to develop their skills.

• Key Factors Influencing Talent Development:

1. Physical Factors – Strength, endurance, agility.
2. Physiological Factors – Recovery, adaptation, workload management.
3. Sociological Factors – Family support, coaching, access to resources.
4. Psychological Factors – Motivation, mental toughness, stress handling.
5. Obstacles – Injuries, burnout, lack of facilities, financial constraints.

B. Sports Training Cycles

Training is structured into different cycles to achieve long-term performance goals:

1. Micro Cycle – Short-term training (3-10 days), focuses on immediate improvement.
2. Meso Cycle – Medium-term (3-6 weeks), works on building endurance, strength, and technique.
3. Macro Cycle – Longest cycle (up to 12 months), includes preparatory, competition, and recovery phases.

Each cycle integrates different intensities and types of training to optimize performance.

C. Development of Strength, Endurance, and Speed

1. Strength : Strength is the ability to exert force against resistance.

Types of Strength

1.1. Maximum Strength – Ability to exert maximum force (e.g., weightlifting).
1.2. Explosive Strength – Ability to exert force quickly (e.g., sprinting, long jump take-off).
1.3. Strength Endurance – Ability to sustain force over time (e.g., wrestling, cycling).

Methods to Develop Strength

1. **Isometric Exercises** – Static exercises where muscle length does not change (e.g., holding a plank, archery).
2. **Isotonic Exercises** – Exercises with visible muscle contraction (e.g., weightlifting, running, jumping).
3. **Isokinetic Exercises** – Performed using machines that adjust resistance (e.g., rowing, swimming).

2. Endurance : Endurance is the ability to sustain physical activity over a prolonged period.

Types of Endurance

1. Basic Endurance – Low-intensity, long-duration activities (e.g., jogging, swimming).
2. General Endurance – Overall ability to resist fatigue in various activities.
3. Specific Endurance – Sport-specific endurance needs (e.g., a boxer vs. a marathon runner).

Methods to Develop Endurance

1. **Continuous Method** – Long, steady-state exercise with no rest (e.g., long-distance running, cycling).
2. **Interval Method** – Alternating high-intensity work and rest periods (e.g., sprint intervals in track and field).
3. **Fartlek Training** – Swedish-origin "speed play," varying intensity throughout (e.g., alternating jogging and sprinting).

3. Speed : Speed is the ability to perform movements quickly.

Types of Speed

1. **Reaction Speed** – Time taken to respond to a stimulus (e.g., sprint start).
2. **Acceleration Speed** – Time taken to reach maximum speed.
3. **Movement Speed** – Performing a single action quickly (e.g., hitting a ball in tennis).
4. **Locomotor Speed** – Sustaining maximum speed for a duration.
5. **Speed Endurance** – Maintaining speed under fatigue.

Methods to Develop Speed

1. **Pace Runs** – Running a set distance at a steady speed (e.g., 800m races).
2. **Acceleration Runs** – Gradual increase in speed, useful for sprinters.

D. Development of Flexibility and Coordinative Abilities

1. Flexibility : Flexibility refers to the range of motion of a joint. It is essential for injury prevention and better performance.

Methods to Develop Flexibility

1. **Slow Stretch and Hold** – Stretching a joint to its maximum and holding (3-8 seconds).
2. **Ballistic Stretching** – Quick, bouncing movements (requires proper warm-up).
3. **Post-Isometric Stretching (PNF)** – Combines passive and isometric stretching for maximum flexibility gains.

2. Coordinative Abilities : Coordinative abilities depend on the central nervous system and enhance movement efficiency, accuracy, and rhythm.

E. Circuit Training

What is Circuit Training?

Circuit training involves performing a series of exercises in a loop (circuit) to develop strength and endurance. It was designed by Adamson and Morgan in 1957.

Characteristics of Circuit Training -

1. Simple and easy-to-learn exercises.
2. Uses medium resistance or body weight.
3. Exercises target different muscle groups.
4. Helps in developing endurance, strength, and overall fitness.

Importance of Circuit Training -

1. Improves cardiovascular fitness.
2. Enhances muscle endurance and strength.
3. Increases VO2 max (oxygen utilization).
4. Promotes whole-body fitness.
5. Key Questions from CBSE Exams & Sample Papers

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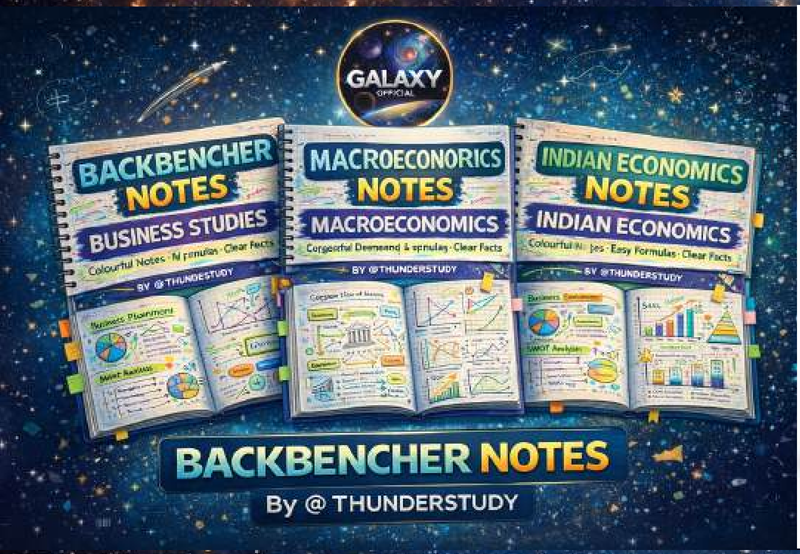

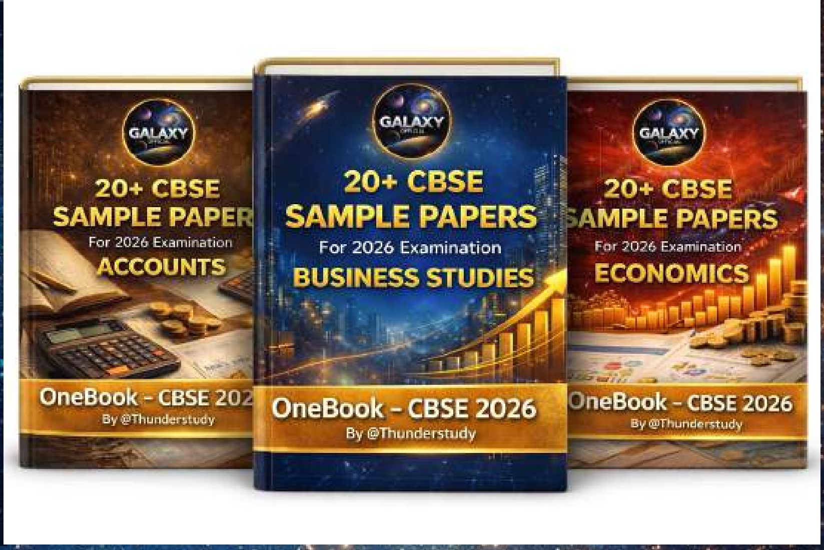

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